Allegra Cuzzocrea

Curriculum Vitae

Department of Physics Ettore Pancini
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Current position

2023–present **PhD, Computational Intelligence**, *University of Naples Federico II*, Naples.
Machine learning and quantum computing

Education

2021–2023 **Master of Computer Engineering**, *University of Salerno*, Salerno).
110/100 cum laude

2018–2021 **Bachelor of Computer Engineering**, *University of Salerno*, Salerno, 110/100 cum laud.
110/100 cum laude

2017–2023 **Bachelor of Piano**, *Conservatoire Giuseppe Martucci of Salerno*, Salerno.
109/100

Conference activities

2025 **Reviewer**, *2025 IEEE Quantum Artificial Intelligence (QAI 2026)*, Nottingham, UK.

2026 **Reviewer and Speaker**, *2026 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2026)*, Maastricht, Netherlands.

2026 **Reviewer**, *2026 IEEE International Joint Conference on Neural Networks (IJCNN 2026)*, Maastricht, Netherlands.

2026 **Reviewer**, *2026 IEEE Conference on Artificial Intelligence (CAI 2026)*, Granada, Spain.

2025 **Staff member and Speaker**, *2025 IEEE Quantum Artificial Intelligence (QAI 2025)*, Naples, Italy.

2025 **Reviewer**, *2025 IEEE International Joint Conference on Neural Networks (IJCNN 2025)*, Rome, Italy.

2025 **Speaker**, *2025 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2025)*, Reims, France.

2025 **Reviewer and Speaker**, *2025 IEEE International Joint Conference on Neural Networks (IJCNN 2025)*, Rome, Italy.

2025 **Speaker**, *Symposium Series o Computational Intelligence (SSCI 2025)*, Trondheim, Norway.

2025 **Participation**, *2025 Workshop on Quantum Computational Intelligence*, Trondheim, Norway.

Editorial activities

Reviewer, *Springer Quantum Machine Learning*.

Reviewer, *IEEE Journal on Selected Areas in Communications*.

Invited talks

2026 **Invited speaker**, *CERN Quantum Technology Initiative Journal Club* , Geneva, Switzerland.
Presented the fundamental concepts of quantum error correction and the latest results in reinforcement learning for online hardware control parameters calibration.

2026 **Invited speaker**, *CERN seminar*, Geneva, Switzerland.
Presented an overview of key challenges in quantum computing and of machine learning applications to quantum computing, highlighting my PhD research and the activities carried out at CERN during my visiting period.

Publications

Conference Proceedings

2026 Giovanni Acampora, Allegra Cuzzocrea, and Autilia Vitiello. An explainable fuzzy logic-based approach to quantum backend selection. In *2026 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pages 1–6, 2026.

2025 Giovanni Acampora, Allegra Cuzzocrea, Marco Lapegna, Roberto Schiattarella, and Autilia Vitiello. Using topology-aware reinforcement learning to synthesize quantum linear reversible circuits. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2025.

2025 Giovanni Acampora, Allegra Cuzzocrea, Marco Lapegna, Roberto Schiattarella, and Autilia Vitiello. Generalizing reinforcement learning-based quantum circuit synthesis across multiple topologies. In *2025 IEEE Symposium for Multidisciplinary Computational Intelligence Incubators (MCII)*, pages 1–7, 2025.

2025 Giovanni Acampora, Allegra Cuzzocrea, Marco Lapegna, Roberto Schiattarella, and Autilia Vitiello. Fuzzy reward-based reinforcement learning for clifford circuit synthesis. In *2025 IEEE International Conference on Fuzzy Systems (FUZZ)*, pages 1–6, 2025.

Submitted Journal Manuscripts

2026 Giovanni Acampora, Allegra Cuzzocrea, and Autilia Vitiello. Qubit2qubit: an interactive and artificial intelligence-based web application for exploring quantum compilation. *Submitted to IEEE Transactions on Education*, 2026. Under review.

2026 Giovanni Acampora, Allegra Cuzzocrea, and Roberto Schiattarella. Optimizing initial qubit layouts for reinforcement learning-based quantum circuit synthesis. *Submitted to IEEE Transactions on Quantum Engineering*, 2026. Under review (second-round reviews received).

2026 Giovanni Acampora, Allegra Cuzzocrea, and Roberto Schiattarella. Machine learning-based extrapolators for zero-noise extrapolation. *Submitted to ACM Transactions on Quantum Computing*, 2026. Under review (first-round reviews received).

Research grants

2025 **EVOQT: EVOlutionary reinforcement learning for Quantum circuit Transpilation**, *IEEE Computational Intelligence Research Student Grant*, first ranked.

Research experience

CERN

01/2026 – 05/2026 ***Reinforcement learning approaches for quantum error correction***.
Developing reinforcement learning-based algorithms for quantum error correction decoders.

Advisor: **Dr. Michele Grossi**, *Coordinator of the Hybrid Quantum Computing Infrastructures and Algorithms competence centre at CERN*, (*IEEE Xplore - Michele Grossi*)

Fraunhofer Institute for Intelligent Analysis and Information Systems (IAIS)

09/2025 – 11/2025 ***EVOQT: EVOlutionary reinforcement learning for Quantum circuit Transpilation***.
Developing evolutionary optimizations to support quantum transpilation algorithms.

Advisor: **Dr. Nico Piatkowski**, *Senior Research Scientist, Fraunhofer Institute for Intelligent Analysis and Information Systems (IAIS)*, (*IEEE Xplore - Nico Piatkowski*)

Position of responsibility

2026 **Research co-supervision of Master's thesis**.
Co-supervised the research activities for a Master's thesis on the systematic analysis of the performance of quantum error correction techniques under noise mismatch scenarios.

2023 **Mentor, Qiskit Fall Fest UNINA**, *University of Naples Federico II*, Naples.
Mentored students during workshops and project activities on quantum computing and Qiskit, and delivered a lecture as part of the event.

Academic achievements & awards

2023 **Excellent academic career** at University of Salerno, Italy

Fellowships

04/2022–10/2022 **Computer engineer**, *A.I. Tech*, Fellowship winner.

Computer skills

Programming Languages	Python, JAVA, C, C++
Frameworks	Qiskit
Web Technologies	HTML, PHP, Javascript
Database	SQL

Certificates

2025 Qiskit Global Summer School - Quantum Excellence
2024 Qiskit Fall Fest Mentor
2024 Qiskit Global Summer School - Quantum Excellence
2024 IBM Quantum Challenge 2024 Achievement
2022 Certificate of Cornell, Maryland, Max-Planck Pre-Doctoral Research School, 2022

Professional memberships

2021–present Student Member, IEEE (Institute of Electrical and Electronics Engineers)